Examiner only

6. (a) State what is meant by the work function, ϕ , of a metal surface. [1]

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(b) Sodium will undergo the photoelectric effect when illuminated by visible light, but zinc requires ultraviolet radiation. **Explain** which material has the greater work function. [1]

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(c) (i) Use Einstein's photoelectric effect equation to show that the maximum wavelength, λ_{max}, for emission is given by the equation: [2]

$$\lambda_{max} = \frac{hc}{\phi}$$

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(ii) A mixture of red ($\lambda = 650\,nm$), green ($\lambda = 550\,nm$) and blue ($\lambda = 450\,nm$) light is incident on a metal surface of work function of 3.7×10^{-19} J. Determine which wavelength or wavelengths of light will be **unable** to release electrons from the metal surface. [2]

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(iii) Explain, in terms of photons, whether or not the intensity of the light will affect the maximum kinetic energy of the released electrons. [2]

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